$$dS = \mu\, dt + \sigma\, dW_t$$

$$E[S(t)] = S(0) + \mu t$$

$$E[S(t)^2] = (S(0) + \mu t)^2 + \sigma^2 t$$

$$E[S(t)^3] = ?$$

$$f(t, S) = S^3 \Rightarrow \frac{\partial f}{\partial t} = 0 \quad \frac{\partial f}{\partial S} = 3S^2 \quad \frac{\partial^2 f}{\partial S^2} = 6S$$

$$dS^3 = df(t, S)$$

$$= \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S}\mu + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2\right)dt + \frac{\partial f}{\partial S}\sigma\, dW_t$$

$$= \left(0 + 3S^2 \cdot \mu + \frac{1}{2} \times 6S\sigma^2\right)dt + \{\cdots\}\, dW_t$$

$$d(E[S(t)^3]) = \left(3E[S(t)^2]\mu + 3E[S(t)]\sigma^2\right)dt$$

$$\int_0^t d(E[S(v)^3]) = \int_0^t 3E[S(v)^2]\mu + 3E[S(v)]\sigma^2\, dv$$

$$= \int_0^t 3\left[(S(0) + \mu v)^2 + \sigma^2 v\right)\mu + 3(S(0) + \mu v)\sigma^2\, dv$$

$$= \int_0^t 3(S(0)+\mu v)^2 \mu \, dv + \int_0^t 3(S(0)+2\sigma^2 \mu v)\, dv$$

$$= \int_0^t 3(S(0)+\mu v)^2 \, d(S(0)+\mu v) + \int_0^t 3S(0)\sigma^2 dv + 6\int_0^t \sigma^2 \mu v \, dv$$

$$= (S(0)+\mu v)^3 \Big|_0^t + 3S(0)\sigma^2 v \Big|_0^t + 3\sigma^2 \mu v^2 \Big|_0^t$$

$$= (S(0)+\mu t)^3 - S(0)^3 + 3S(0)\sigma^2 t - 0 + 3\sigma^2 \mu t^2 - 0$$

$$\int_0^t d\left[E[S(v)^3]\right] = (S(0)+\mu t)^3 - S(0)^3 + 3S(0)\sigma^2 t + 3\sigma^2 \mu t^2$$

$$E[S(t)^3] - S(0)^3 = (S(0)+\mu t)^3 - S(0)^3 + 3S(0)\sigma^2 t + 3\sigma^2 \mu t^2$$